

Tarun Maity

POSTDOCTORAL RESEARCHER · COMPUTATIONAL BIOPHYSICS & MOLECULAR MODELING

RIKEN Cluster for Pioneering Research (CPR), Wako, Saitama, Japan

+91-7384688688 · tarunrkmrc95@gmail.com · [LinkedIn](#) · [Google Scholar](#) · [ResearchGate](#)

RESEARCH PROFILE

I am a computational physicist and postdoctoral researcher at RIKEN CPR developing machine-learning-enhanced molecular dynamics models for soft matter and biological polymer systems. My doctoral research at IISc Bengaluru established multi-scale molecular simulation frameworks to study dendrimer-based systems across three application domains: energy storage (supercapacitors), heavy-metal remediation (uranyl extraction), and targeted drug delivery (siRNA complexation and cytotoxicity prediction). I combine physics-based simulation (MD, MC, DFT) with data-driven methods (neural networks, CNNs) to extract predictive structure-property relationships. My work has appeared in *Physical Review Applied*, *Langmuir*, and *ACS Omega*, and I hold top All-India competitive examination ranks (JEST: 8, JAM: 19, CSIR-UGC NET JRF: 25, GATE: 33).

RESEARCH INTERESTS

Molecular Dynamics & Coarse-Grained Simulation · Machine Learning Force Fields · Dendrimer & Polymer Physics · Energy Storage Materials · Nuclear Waste Remediation · Drug Delivery & Nanotoxicology · Soft Matter & Biophysics

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher 2024 – Present
RIKEN Cluster for Pioneering Research (CPR), Theoretical Molecular Science Lab Wako, Japan

- Developing ML-enhanced coarse-grained force fields for intrinsically disordered protein and polymer systems using extensive MD simulations, in collaboration with Prof. Yuji Sugita.
- Applying enhanced sampling techniques (replica exchange, metadynamics) to characterize conformational landscapes of disordered biopolymers.

Technical R&D Specialist Aug 2023 – Apr 2024
Accenture, Life Sciences Division India

- Curated the first publicly available, physicochemical-feature-annotated dataset for PAMAM dendrimer cytotoxicity, enabling reproducible benchmarking of ML toxicity models.
- Developed and benchmarked six ML algorithms (Random Forest, XGBoost, SVM, ANN) for cytotoxicity prediction in low-data regimes; results published in *ACS Omega* (2024).
- Delivered internal workshops on data-driven drug design to a team of 12 R&D engineers.

EDUCATION

Ph.D., Physics 2018 – 2024
Indian Institute of Science (IISc) Bengaluru, India

- **Thesis:** *Exploring Applications of Dendrimers in Energy Storage, Uranyl Extraction, and Drug Delivery through Molecular Modeling*
- **Supervisor:** Prof. Prabal K. Maiti
- Developed original multi-scale simulation protocols combining AMBER/GROMACS with in-house Python analysis pipelines; introduced DFT-calibrated force field parameters for graphene-PAMAM nanocomposites.

M.Sc., Physics 2016 – 2018

- Thesis/Project: (Add title if applicable) · Graduated July 2018

B.Sc. (Hons.), Physics

2013 – 2016

Ramakrishna Mission Residential College (Autonomous)

Narendrapur, Kolkata, India

- Minors: Mathematics, Statistics · Graduated July 2016

PUBLICATIONS

6 peer-reviewed publications; first author on all.

2025

1. **Maity, T.**, Kumar, Y., & Maiti, P. K. (2025). Complexation of siRNA with galactose-functionalized dendrimers: PAMAM vs. PETIM. *Langmuir*. doi:10.1021/acs.langmuir.5c03613
2. **Maity, T.**, Kumar, Y., Singha Deb, A. K., Sheikh, M. A., & Maiti, P. K. (2025). Enhanced and efficient extraction of uranyl ions from aqueous waste using graphene/CNT-PAMAM nanocomposites. *Langmuir*, **41**(22), 13811–13823. doi:10.1021/acs.langmuir.5c00280

2024

3. **Maity, T.**, Balachandran, A. K., Krishnamurthy, L. P., Nagar, K. L., Upadhyayula, R. S., Sengupta, S., & Maiti, P. K. (2024). Data-driven approaches to predict dendrimer cytotoxicity. *ACS Omega*, **9**(23), 24899–24906. doi:10.1021/acsomega.4c01775

2023

4. **Maity, T.**, Aggarwal, A., Dasgupta, S., Velachi, V., Singha Deb, A. K., Ali, S. M., & Maiti, P. K. (2023). Efficient removal of uranyl ions using PAMAM dendrimer: Simulation and experiment. *Langmuir*, **39**(19), 6794–6802. doi:10.1021/acs.langmuir.3c00332

2022

5. **Maity, T.**, Gosika, M., Pascal, T. A., & Maiti, P. K. (2022). Molecular insights into the physics of poly(amidoamine)-dendrimer-based supercapacitors. *Physical Review Applied*, **18**(5), 054031. doi:10.1103/PhysRevApplied.18.054031

FELLOWSHIPS, AWARDS & DISTINCTIONS

- 2013–2018 **INSPIRE Scholarship**, Dept. of Science and Technology, Govt. of India — competitive national fellowship awarded to top science undergraduates and postgraduates (approx. 10,000 recipients/year from ~500,000 applicants).
- 2016 **JEST Physics 2016 — All-India Rank 8** (~15,000 participants)
- 2016 **JAM Physics 2016 — All-India Rank 19** (~15,000 participants)
- 2016 **NGPE (IAPT) 2016 — 99th Percentile**, National Graduate Physics Examination (~20,000 participants)
- 2018 **CSIR-UGC NET JRF 2018 — All-India Rank 25** (~15,000 participants)
- 2018 **GATE Physics 2018 — All-India Rank 33** (~15,000 participants)

TALKS & PRESENTATIONS**Invited/Contributed Talks**

- **CCP2023** (Conference on Computational Physics), Kobe, Japan, 2023. (Title: add)
- **Statistical Biological Physics: from Single Molecule to Cell**, ICTS Bangalore, 2022. (Title: add)
- **Inhouse Symposium**, IISc Bangalore, 2023.

Poster Presentations

- **RARE 2024** (Recent Advances in Modeling Rare Events), Khajuraho, India, 2024.
- **ACS March Meeting 2021**.

- **Inhouse Symposium**, IISc Bangalore, 2020.

TEACHING & MENTORING EXPERIENCE

IISc, 2021	Teaching Assistant, <i>Computational Physics</i> (graduate-level; conducted tutorials and graded assignments for ~30 students)
IISc, 2020	Teaching Assistant, <i>Mathematical Methods for Physics</i> (graduate-level)
2018–present	Science mentor for secondary-school students (ongoing, voluntary)

TECHNICAL SKILLS

Simulation Methods	Molecular Dynamics (AMBER, GROMACS, LAMMPS), Monte Carlo, DFT (GAUSSIAN, Quantum ESPRESSO, VOTCA-CTP), Enhanced Sampling (REMD, Metadynamics)
ML / Data Science	TensorFlow, Scikit-Learn, PyTorch (basic), CNNs, Random Forests, XG-Boost, NLP; experienced in small-data ML regimes
Programming	Python (NumPy, Pandas, Matplotlib, SciPy), Fortran, Bash/Shell, VMD Tcl, Mathematica
Visualization	VMD, PyMOL, Xmgrace, Gnuplot, Origin, Inkscape
HPC / DevOps	Cluster job scheduling (PBS/SLURM), cluster maintenance, software compilation; Google Cloud Platform, Git, Jupyter, VS Code
Writing	LaTeX, Mendeley; experienced in drafting manuscripts, grant summaries, and technical reports
Languages	Bengali (native), English (professional), Hindi (fluent)

REFERENCES

Prof. Yuji Sugita

Chief Scientist
RIKEN Cluster for Pioneering Research (CPR)
Wako, Saitama, Japan
Email: available upon request
Phone: available upon request

Prof. Prabal K. Maiti

Professor, Dept. of Physics
Indian Institute of Science (IISc)
Bengaluru 560012, India
Email: available upon request
Phone: available upon request

A third reference (industry/cross-disciplinary) can be provided upon request.